Handling Missing Data in ETL Assignment

SECTION A - Theoretical Questions

Question 1. What are the most common reasons for missing data in ETL pipelines?

- Data extraction errors: Network timeouts, API failures, or connection drops during the extraction phase leading to incomplete records.
- Source system changes: Schema updates, deprecated columns, or legacy system migrations where fields were added or removed.
- User omission: Optional form fields left blank by users during data entry (e.g., optional profile details).
- System bugs or formatting issues: Data corruption during parsing (e.g., mismatch in delimiters, encoding errors, or type conversion failures).

Question 2. Why is blindly deleting rows with missing values considered a bad practice in ETL?

- Information loss: It discards valuable data present in other columns of the same row, reducing the overall dataset size.
- Introduction of bias: If missingness is not completely random (e.g., high-income earners skipping their income field), deleting rows skews the distribution and compromises statistical validity.
- Reduced model performance: Smaller training datasets lead to less robust machine learning models and inaccurate analytics.

Question 3. Explain the difference between listwise deletion and column deletion.

- Listwise Deletion: Removes entire rows (records) that contain one or more missing values across specified columns.
  Appropriate scenario: When a very small fraction of rows have missing values, and discarding them does not introduce sample bias.
- Column Deletion: Removes an entire column (feature/attribute) from the dataset.
  Appropriate scenario: When a specific column has a massive proportion of missing values (e.g., >80% missing) and offers little to no analytical value.

Question 4. Why is median imputation preferred over mean imputation for skewed data such as income?

- Robustness to outliers: Income distributions are typically right-skewed (a few high earners pull the mean upward). The mean is sensitive to these extreme values, whereas the median represents the middle value and provides a more realistic measure of central tendency for skewed data.

Question 5. What is forward fill and in what type of dataset is it most useful?

- Forward fill: A technique that propagates the last observed valid value forward to fill subsequent missing values until a new valid value is encountered.
- Usefulness: Most useful in ordered or sequential datasets such as time-series data (e.g., stock prices, sensor readings, or daily transactions) where values tend to persist until an explicit change occurs.

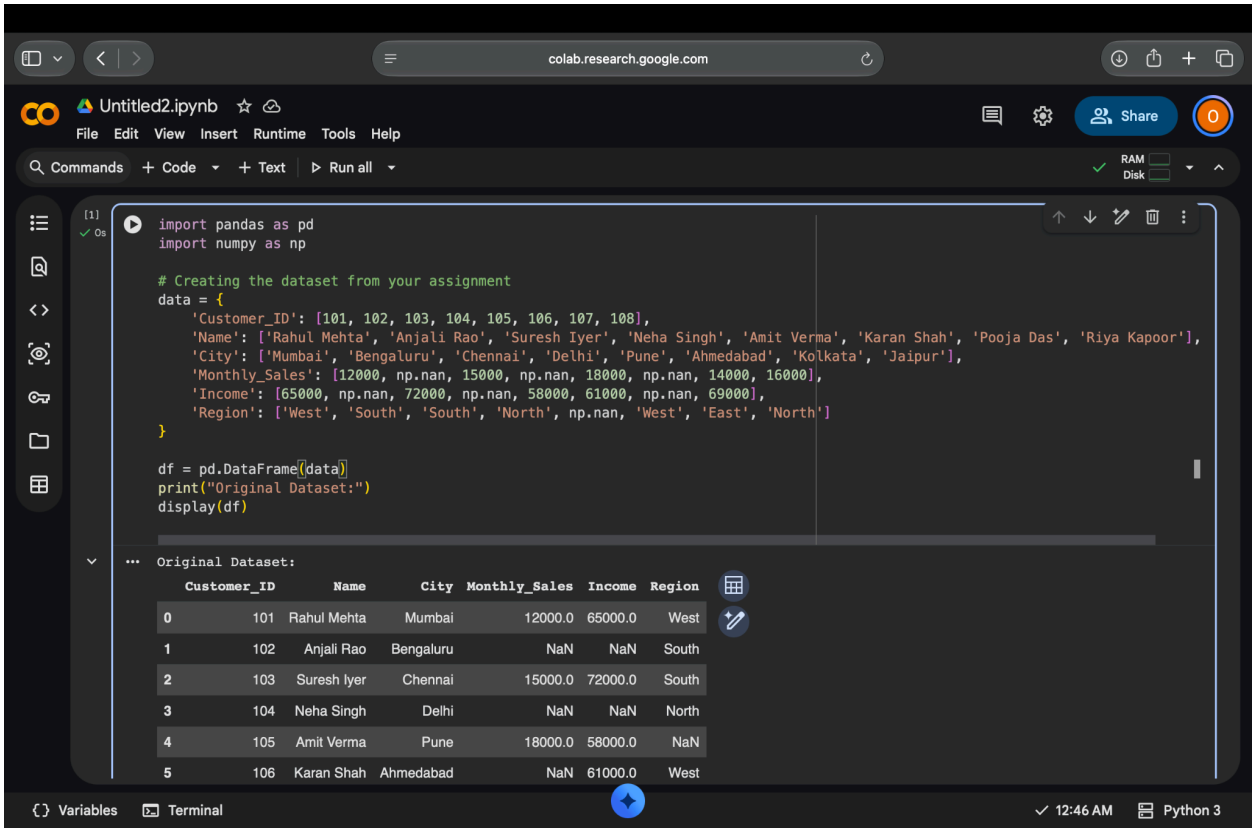
Question 6. Why should flagging missing values be done before imputation in an ETL workflow?

- Preserving missingness information: The fact that a value is missing can itself carry predictive signal (Missing Not At Random or Missing At Random). Creating a binary flag before imputation ensures downstream models can capture this pattern rather than treating imputed values as real observations.

Question 7. Consider a scenario where income is missing for many customers. How can this missingness itself provide business insights?

- Behavioral segmentation: Customers who withhold their income might share common traits (e.g., privacy-conscious shoppers, lower willingness to pay, or specific demographic segments).
- Correlation with engagement: Non-disclosure of income can be correlated with lower checkout conversion rates, specific product categories browsed, or churn risk, which can be leveraged as a distinct predictor feature in predictive analytics.

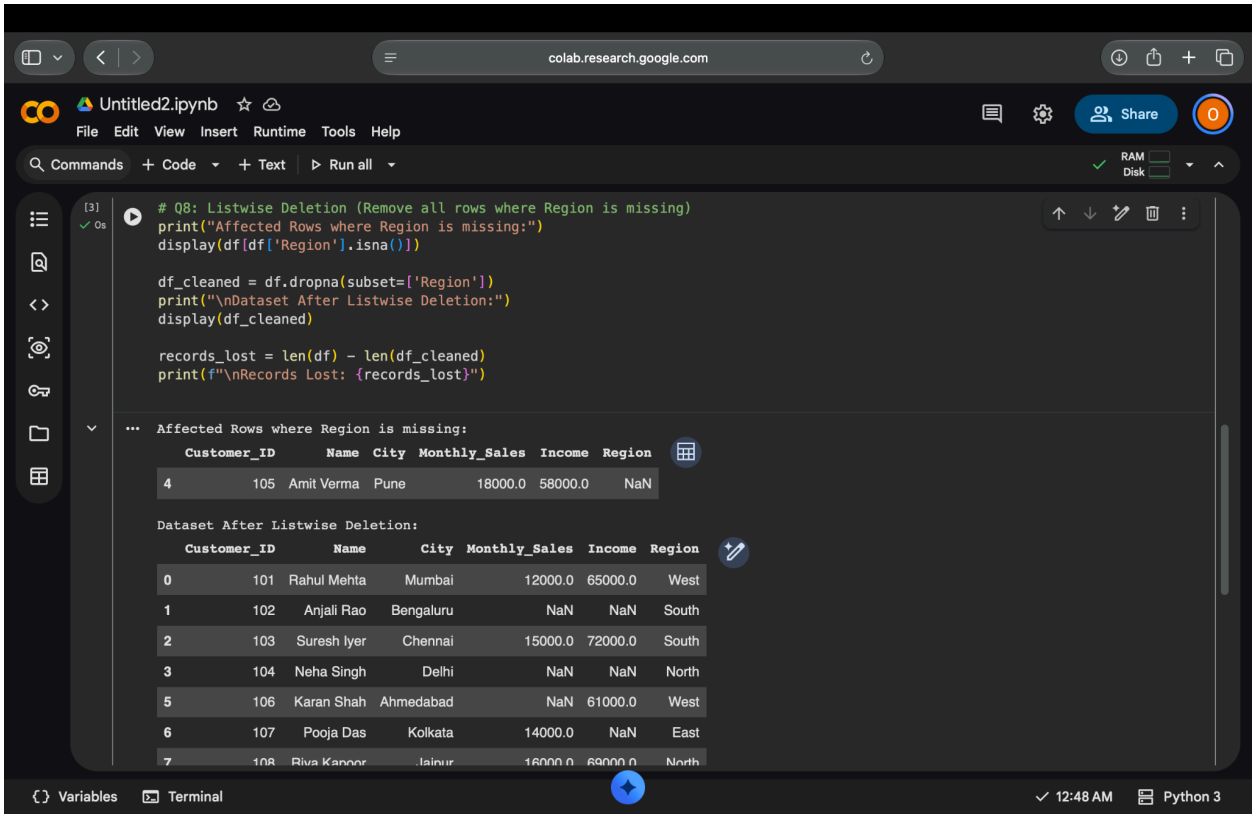
SECTION B - Practical Questions : DATASET :-

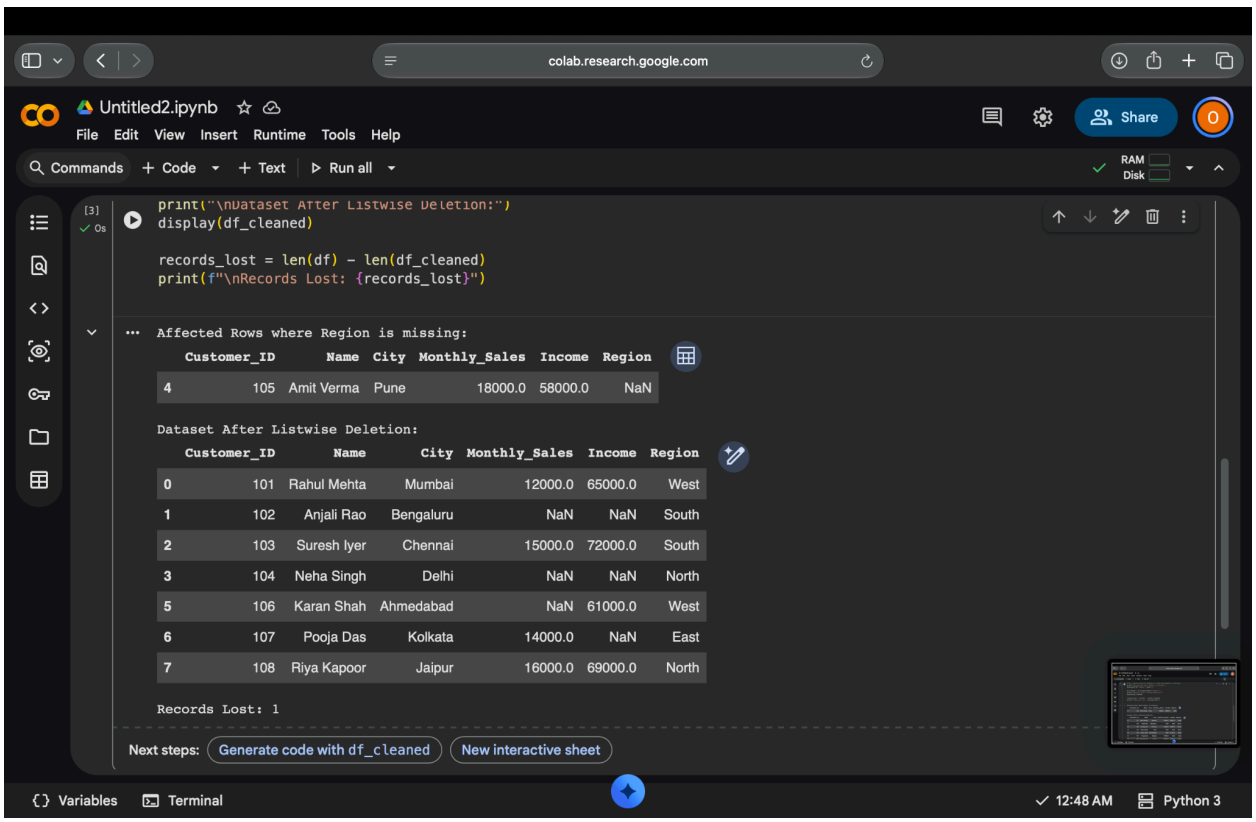


Question 8. Listwise Deletion

- Remove all rows where the **Region** is missing.
- **Tasks:**
  . Identify affected rows
  .Show the dataset after deletion
  . Mention how many records were lost

Answer :


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Q8: Listwise Deletion (Remove all rows where Region is missing)
print("Affected Rows where Region is missing:")
display(df[df['Region'].isna()])
df_cleaned = df.dropna(subset=['Region'])
print("\nDataset After Listwise Deletion:")
display(df_cleaned)
records_lost = len(df) - len(df_cleaned)
print(f"\nRecords Lost: {records_lost}")
Affected Rows where Region is missing:
Customer_ID Name City Monthly_Sales Income Region
4 105 Amit Verma Pune 18000.0 58000.0 NaN
Dataset After Listwise Deletion:
Customer_ID Name City Monthly_Sales Income Region
0 101 Rahul Mehta Mumbai 12000.0 65000.0 West
1 102 Anjali Rao Bengaluru NaN NaN South
2 103 Suresh Iyer Chennai 15000.0 72000.0 South
3 104 Neha Singh Delhi NaN NaN North
5 106 Karan Shah Ahmedabad NaN 61000.0 West
6 107 Pooja Das Kolkata 14000.0 NaN East
Variables Terminal
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print("\nDataset After Listwise Deletion:")
display(df_cleaned)
records_lost = len(df) - len(df_cleaned)
print(f"\nRecords Lost: {records_lost}")
Affected Rows where Region is missing:
Customer_ID Name City Monthly_Sales Income Region
4 105 Amit Verma Pune 18000.0 58000.0 NaN
Dataset After Listwise Deletion:
Customer_ID Name City Monthly_Sales Income Region
0 101 Rahul Mehta Mumbai 12000.0 65000.0 West
1 102 Anjali Rao Bengaluru NaN NaN South
2 103 Suresh Iyer Chennai 15000.0 72000.0 South
3 104 Neha Singh Delhi NaN NaN North
5 106 Karan Shah Ahmedabad NaN 61000.0 West
6 107 Pooja Das Kolkata 14000.0 NaN East
7 108 Riya Kapoor Jaipur 16000.0 69000.0 North
Records Lost: 1
Next steps: Generate code with df_cleaned New interactive sheet
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Q9. Imputation

.Handle missing values in **Monthly_Sales** using:

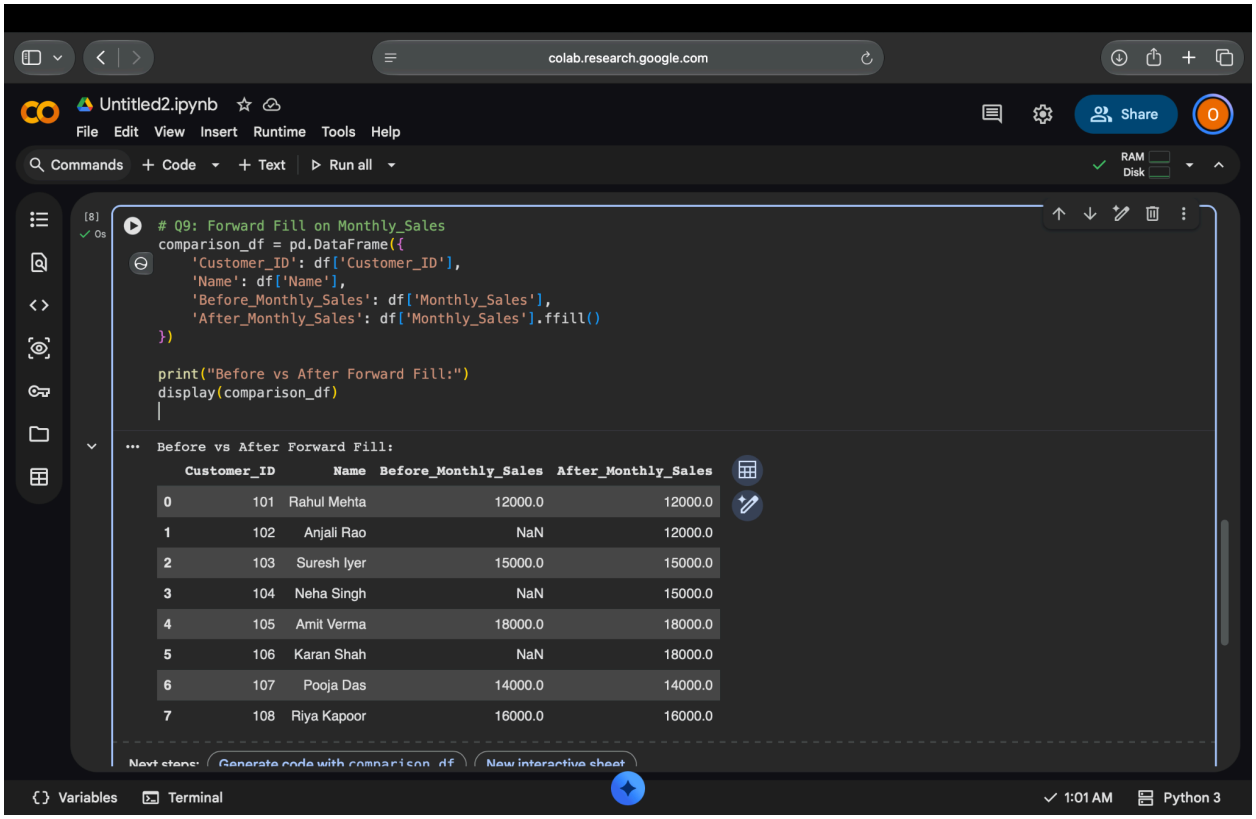
–Forward Fill

Tasks:

.Apply forward fill Show before vs after values

.Explain why forward fill is suitable here

Answer:



	Customer_ID	Name	Before_Monthly_Sales	After_Monthly_Sales
0	101	Rahul Mehta	12000.0	12000.0
1	102	Anjali Rao	NaN	12000.0
2	103	Suresh Iyer	15000.0	15000.0
3	104	Neha Singh	NaN	15000.0
4	105	Amit Verma	18000.0	18000.0
5	106	Karan Shah	NaN	18000.0
6	107	Pooja Das	14000.0	14000.0
7	108	Riya Kapoor	16000.0	16000.0

Q10. Flagging Missing Data

Create a **flag column** for missing Income.

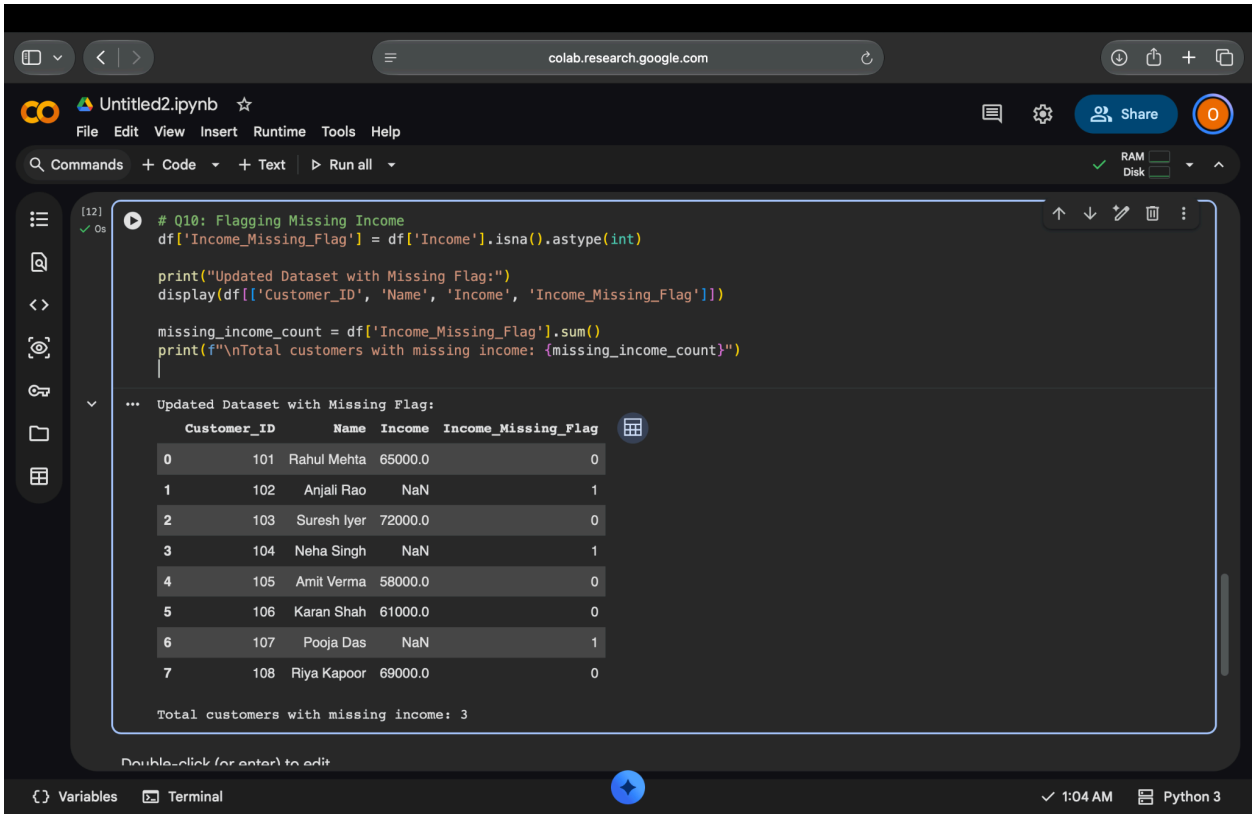
Tasks:

.Create Income_Missing_Flag (0 = present, 1 = missing)

.Show updated dataset

.Count how many customers have missing income

Answer :


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Q10: Flagging Missing Income
df['Income_Missing_Flag'] = df['Income'].isna().astype(int)
print("Updated Dataset with Missing Flag:")
display(df[['Customer_ID', 'Name', 'Income', 'Income_Missing_Flag']])
missing_income_count = df['Income_Missing_Flag'].sum()
print(f"\nTotal customers with missing income: {missing_income_count}")
Updated Dataset with Missing Flag:
Customer_ID Name Income Income_Missing_Flag
0 101 Rahul Mehta 65000.0 0
1 102 Anjali Rao NaN 1
2 103 Suresh Iyer 72000.0 0
3 104 Neha Singh NaN 1
4 105 Amit Verma 58000.0 0
5 106 Karan Shah 61000.0 0
6 107 Pooja Das NaN 1
7 108 Riya Kapoor 69000.0 0
Total customers with missing income: 3
Double-click (or enter) to edit
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